

ORIGINAL ARTICLE

When Does Low-Light Image Restoration Help Industrial Anomaly Detection?

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ABSTRACT

Low illumination reduces both the visibility and the reliability of industrial image inspection. We study the question of when low-light restoration actually helps downstream industrial anomaly detection, rather than assuming that better visibility implies better inspectability. We first introduce ICRA-MIRNet, an illumination-conditioned residual adapter wrapped around a pretrained MIRNet-v2 backbone and fine-tuned on normal industrial training images under a deterministic camera-inspired low-light formation model. We then fix an ImageNet-pretrained PatchCore probe [1] and evaluate clear, darkened, and restored inputs across all 15 MVTec AD categories at mild, training-range, and severe degradation. Restoration improves training-range anomaly AUROC from 0.800 to 0.938 (+0.138), but the gain shrinks to +0.018 under mild degradation and +0.019 under severe degradation, while the generic MIRNet-v2 baseline actually degrades mild performance (−0.016). A label-free restore-or-bypass gate learned only from normal detector-consistency utility cannot reliably select when to restore: it underperforms always-restore on two of three severities, with a mean Gate-minus-Always margin of −0.028. The results show that restoration benefit is condition- and method-dependent and that normal-domain consistency utility is an inadequate proxy for anomaly-discrimination utility. Controlled synthetic evidence cannot be extended to real night-time factory deployments.

1 | Introduction

Industrial anomaly inspection is typically developed from well-exposed training images, whereas installed cameras may be subject to illumination gradients, photon noise, and read noise. Image enhancement can improve visibility, but a restoration model that alters defect evidence may increase rather than reduce inspection risk. Consequently, restoration fidelity and downstream anomaly discrimination must be evaluated together.

MIRNet-v2 preserves high-resolution representations while exchanging contextual information across multiple resolutions, and it supports several restoration tasks including low-light enhancement [2]. Its generic output, however, is not conditioned on the spatial illumination pattern of a specific industrial image. We therefore wrap its output residual with a learnable illumination-conditioned gate that retains the pretrained mapping when the

gate approaches one and permits spatially selective correction after low-memory adaptation.

Our central question is not whether restoration improves visibility, which is expected, but whether it improves downstream anomaly detection, and under which conditions it does so. We use MVTec AD [3] as the primary industrial benchmark because it provides normal training images and normal/anomalous test images across object and texture categories. A frozen PatchCore detector [1] serves as the downstream probe and is applied verbatim to clear, synthetically darkened, and restored images so that only the input domain changes. Low-light observations are generated deterministically from the original clear images under three severity intervals.

The intended contributions are:

- an illumination-conditioned residual formulation that adapts a pretrained MIRNet-v2 model on a single 8-GB GPU;
- a leakage-controlled protocol that trains only on normal images and jointly reports PSNR, SSIM, and PatchCore class-macro anomaly AUROC;
- a controlled three-severity analysis that identifies when restoration helps, when it is marginal, and when it harms;
- a label-free restore-or-bypass gate and an explicit audit showing that normal-domain detector-consistency utility is an inadequate surrogate for anomaly-discrimination utility.

2 | Related Work

2.1 | Low-Light Image Enhancement

The LOL dataset introduced paired low- and normal-light images for supervised enhancement [4]. MIRNet-v2 subsequently provided a multiscale architecture for restoration and enhancement with high-resolution feature preservation [2]. Our work does not propose a replacement restoration backbone; it studies whether a lightweight illumination-conditioned residual adapter can preserve industrial defect evidence under a fixed low-light protocol, and whether that preservation translates to downstream inspectability.

2.2 | Industrial Anomaly Inspection

MVTec AD contains normal training images and normal/anomalous test images across object and texture categories [3]. Memory-bank detectors such as PatchCore achieve strong reconstruction-free unsupervised detection by matching test patches against a coreset of training features [1]. We use image-level anomaly discrimination as the downstream probe. The probe is deliberately frozen across clear, darkened, and restored inputs so that only the image domain changes.

3 | Method

3.1 | Deterministic Low-Light Formation

Let $\mathbf{y} \in [0, 1]^{3 \times H \times W}$ denote a clear industrial image. A seeded degradation operator produces

$$\mathbf{x} = \text{clip}(\text{Poisson}(255 \mathbf{y}^\gamma \odot \mathbf{m}) / 255 + \epsilon, 0, 1), \quad (1)$$

where γ is sampled from a fixed interval, $\mathbf{m}$ is a smooth spatial illumination plane, and ϵ is zero-mean Gaussian read noise. For the training-range protocol, $\gamma \sim \mathcal{U}(2, 4)$ and the read-noise standard deviation is $\mathcal{U}(0, 0.02)$. The mild interval uses $\gamma \sim \mathcal{U}(1.2, 2.0)$ and read noise $\mathcal{U}(0, 0.01)$, and the severe interval uses $\gamma \sim \mathcal{U}(4.0, 5.0)$ and read noise $\mathcal{U}(0.02, 0.05)$. The illumination plane uses horizontal and vertical slopes sampled independently from $\mathcal{U}(-0.30, 0.30)$ and an offset from $\mathcal{U}(0.55, 0.85)$, followed by clipping to $[0.35, 1]$. Poisson noise is sampled at a 255-count scale. The degradation seed is the global seed plus the image index, so each indexed pair is reproducible.

3.2 | Illumination-Conditioned Residual Adapter

Let $B(\mathbf{x})$ be the pretrained MIRNet-v2 output and

$$\ell(\mathbf{x}) = 0.299\mathbf{x}_R + 0.587\mathbf{x}_G + 0.114\mathbf{x}_B \quad (2)$$

be luminance. A two-layer convolutional network predicts a spatial gate

$$\mathbf{g} = \sigma(G(1 - \ell(\mathbf{x}))). \quad (3)$$

The final restoration is

$$\hat{\mathbf{y}} = \mathbf{x} + \mathbf{g} \odot (B(\mathbf{x}) - \mathbf{x}). \quad (4)$$

The final gate layer is initialized with zero weights and bias eight, so the initial gate is close to one and Eq. (4) initially approximates the pretrained backbone.

3.3 | Optimization

The training loss is

$$\mathcal{L} = \|\hat{\mathbf{y}} - \mathbf{y}\|_1 + \lambda_v \|\nabla \hat{\mathbf{y}} - \nabla \mathbf{y}\|_1 + \lambda_g \|\mathbf{g} - (1 - \ell(\mathbf{y}))\|_1, \quad (5)$$

with $\lambda_v = 0.2$ and $\lambda_g = 0.05$. Training uses only the normal train split and never touches test images or anomaly labels.

3.4 | Frozen Downstream Anomaly Probe

For each category c , we use the PatchCore detector [1] trained on clear normal training images only. A Wide-ResNet-50-2 backbone extracts local patch features, a greedy coreset subsampling constructs a compact memory bank, and the anomaly score of a test patch is the squared nearest-neighbor distance to that memory bank. Image-level scores are pooled by the maximum over patches. The detector is frozen after training on clear images and is applied unchanged to clear, darkened, and restored test images. AUROC is computed within each category and macro-averaged across categories.

4 | Experiments

4.1 | Datasets and Protocol

The local MVTec AD copy contains 3,629 normal training images and 1,725 test images across 15 categories. The restoration model sees only the training/normal split; test images never participate in restoration optimization or PatchCore memory-bank construction. The main model processes 12,000 micro-batches using 128×128 crops, batch size two, four-micro-batch gradient accumulation, automatic mixed precision, and a fixed seed. Each indexed training image receives a deterministic crop and degradation under the global seed. Full-resolution restoration uses overlapping 128×128 tiles with 16-pixel overlap.

4.2 | Metrics

Restoration fidelity is measured by image-pooled PSNR and SSIM [5]. Inspection is measured by image-level AUROC

TABLE 1 | MVTEC AD restoration fidelity and downstream PatchCore macro AUROC for the training-range protocol. ICRA-MIRNet reports mean $\pm$ sample standard deviation over three seeds; the LOL-pretrained baseline and component ablations use a single seed.

Method	PSNR $\uparrow$	SSIM $\uparrow$	Dark AUROC	Restored AUROC
LOL MIRNet-v2	18.168	0.6164	0.8000	0.8987
Fixed-gate fine-tuning	28.581	0.8960	0.8000	0.9316
ICRA w/o $\mathcal{L}_V$	28.744	0.8906	0.8000	0.9338
ICRA-MIRNet	28.472 $\pm$ 0.243	0.8946 $\pm$ 0.0015	0.8000	0.9378

TABLE 2 | Class-macro PatchCore anomaly AUROC by degradation severity and input domain. Higher is better; all rows use the frozen PatchCore detectors.

Severity	Dark	MIRNet-v2	ICRA-MIRNet	Δ ICRA–Dark
mild	0.9450	0.9290	0.9629	+0.0179
training-range	0.8000	0.8987	0.9378	+0.1378
severe	0.7569	0.7758	0.7758	+0.0190

TABLE 3 | Restoration fidelity of ICRA-MIRNet across degradation severity intervals. PSNR and SSIM are image-pooled over the restored test images. The downstream anomaly-discrimination effect is reported separately in Table 2 using the frozen class-macro PatchCore detectors.

Input domain	PSNR $\uparrow$	SSIM $\uparrow$
Mild	21.514	0.8640
Training-range	28.472 $\pm$ 0.243	0.8946 $\pm$ 0.0015
Severe	21.466	0.6521

within each category, then macro-averaged without category-size weighting for clear, darkened, and restored inputs. Runtime is the image-pooled restoration latency and peak allocated CUDA memory is recorded for the evaluation phase only. Main-model results are the arithmetic mean and sample standard deviation over three seeds.

4.3 | Baselines and Ablations

The audited matrix contains: (i) the official LOL-pretrained MIRNet-v2 without industrial adaptation; (ii) backbone fine-tuning with a fixed gate $g = 1$; (iii) ICRA-MIRNet without the gradient loss; and (iv) the complete ICRA-MIRNet. Mild and severe degradation intervals test sensitivity outside the training range.

4.4 | Label-Free Restore-or-Bypass Gate

We additionally test whether a policy learned without anomaly labels can decide, per image, whether to restore or bypass. On normal training images we define a detector-consistency utility $U = |s_{\text{dark}} - s_{\text{clear}}| - |s_{\text{restored}} - s_{\text{clear}}|$ and label an image as “restore” when $U > 0$. A 23-dimensional image-only feature vector (luminance percentiles, contrast, block statistics, gradient statistics, and a noise proxy) is fed to a standardized logistic regression. The gate is frozen and applied image-by-image to the full test set. The gate is not degenerate: it restores about 63% of mild, 82% of training-range, and 52% of severe test samples. However, this selectivity does not improve downstream discrimination relative

to always-restore. The repaired macro AUROC for the gate is 0.9636 (mild, +0.0007 vs. always-restore), 0.8809 (training-range, -0.0569), and 0.7492 (severe, -0.0266), with a mean Gate-minus-Always margin of -0.028 . Choosing to restore is not equivalent to choosing to improve anomaly detection.

5 | Reproducibility and Evidence Boundaries

The industrial runner records configuration, checkpoints, training progress, per-image evaluation progress, per-category metrics, and final JSON output. A separate validator rejects missing categories and non-finite metrics before producing mean $\pm$ standard-deviation tables. All low-light inputs in this study are synthetic. Neither MVTEC AD nor any available cross-dataset corpus is a paired night-time industrial dataset, so the experiments establish controlled degradation robustness but cannot establish performance in a real dark factory. A real paired or exposure-bracketed industrial test set remains necessary before any deployment claim.

6 | Conclusion

We presented an illumination-conditioned residual adapter and an audited protocol that evaluates low-light restoration and downstream industrial anomaly discrimination together. Restoration improves training-range PatchCore AUROC by 0.138, but the gain is only 0.018 under mild degradation and 0.019 under severe degradation, and a generic restorer can even hurt the mild regime. A label-free gate cannot reliably identify when restoration is useful. The mixed direction across degradation intensity, restoration method, and policy cautions against treating restoration visibility as a proxy for inspection reliability. Future work must add real exposure variation and an operational defect detector.

Author Contributions

Jin Huang conceived the study, implemented the restoration method and evaluation protocol, and wrote the manuscript. Qisen Gao contributed to the experimental design and the analysis of downstream anomaly detection results. Both authors approved the final manuscript. This manuscript is submitted solely to the target journal and is not under consideration elsewhere.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The derived results tables and the audited per-category records are available from the corresponding author on reasonable request. MVTEC AD is publicly available from its original authors.



FIGURE 1 | One MVtec AD bottle/broken_large sample under training-range degradation. The clear image (left), the deterministic low-light observation (middle), and the ICRA-MIRNet restoration (right) share the same input content but lead to different PatchCore scores.

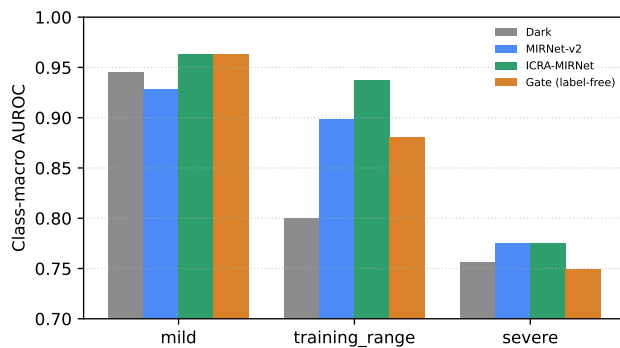


FIGURE 2 | Class-macro PatchCore AUROC by severity and input domain. Restoration helps most in training-range, is marginal in mild/severe, and the label-free gate cannot beat always-restore on two of three severities.

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